

Internship portfolio report

1. Description of the company and role

1.1 Description of the company:

Lely (officially Lely Industries N.V.) is a Dutch, family-owned agricultural technology company founded in 1948 by brothers Cornelis and Arij van der Lely in Maassluis, the Netherlands. It has evolved into a global leader in automated solutions for dairy farming, focusing on innovation to make farm work easier, more efficient, and more sustainable. The company operates in more than 50 countries and has around 4000 employees worldwide.

1.2 Vision:

Lely's vision is centered on creating a sustainable, profitable, and enjoyable future in farming. It aims to lead agricultural innovation by developing technologies that improve animal welfare, reduce environmental impact, enhance dairy quality, and promote farmer prosperity.

This slogan and way of working were always mentioned to me, and there weren't situations where this company-wide view wasn't truthful; in almost every employee, this was seen. And for my work, it had a real impact; speaking with actual farmers, seeing their passion for a sustainable farming future really drove me to keep doing a better job for the sake of everyone. Having good motives to work for and strive for always helped.

1.3 History:

A short list of Lely's biggest accomplishments and products throughout the years:

- 1948: Founded by Cornelis and Arij van der Lely with the goal of making life easier for farmers through innovation.
- Initially developed simple agricultural tools; over time it expanded into sophisticated automated systems.
- 1992: Debut of the Lely Astronaut, one of the first commercial robotic milking systems, which revolutionized dairy farming. I will be working on the iterations that came after the original one, which is, of course, more advanced in every way, but this first iteration was still a first of its kind and symbolized true innovation of its time.
- Lely has since grown its range to include feeding, barn cleaning and herd management automation.
- Today it holds 1,500+ active patents and maintains a massive global presence with dedicated Lely Centers supporting farmers locally.

At Lely, inside and outside the office, regarding the work culture, you can feel the company has a genuine engineering-first and IP-protective based culture with that many patents under their belt. This shaped my perspective even before I came to the company. Innovation breeds growth.

1.4 Products:

Lely develops and manufactures innovative automated and digital solutions for dairy farms, including:

- Milking robots (e.g., Lely Astronaut A5 series) enabling cows to be milked on demand.
- Feeding systems (e.g., Vector, Juno Max) optimizing feeding routines and nutrition.
- Manure and barn handling systems (e.g., Discovery Collector C2).
- Monitoring and management systems (digital tools for herd and barn oversight).
- Services and advice on farm planning and efficient operation.

1.5 Market and target group:

- **Market:** Global agricultural machinery and technology market with emphasis on dairy automation. Lely's solutions are used in more than 50 countries across Europe, the Americas, Asia and beyond.
- **Target Group:** Primarily dairy farmers and livestock producers seeking to increase efficiency, productivity, and sustainability through automation and data-driven farm management. This includes small family farms and larger commercial dairy operations. Lely also serves through a network of Lely Centers, offering local sales, implementation, and after-sales support.

1.6 My project and role in the company:

My specific role in the company and the project I will be mainly working on is called the Beaver project, formerly known as Milk Interval Optimization (MIO). I will work in the company's AI and data analysis department, which has about 40 people. I will be working in the Milk Settings Optimization (MSO) team, which has 4 other people besides me, and I will be the first intern to fill this position for this period. The MSO team has a few other projects underway, but as I stated, I will be mostly working on the Beaver project.

1.7 Beaver project:

The beaver project was created to support the Horizon application, which is found on the Astronaut machines made by Lely. Currently, farmers must manually configure when dairy cows are to be milked on an outdated, confusing page used for milk access. If farmers continue to manually control when the cows are milked, there is bound to be variation and performance loss. The solution that the Beaver project brings is fully automated milk access timing at the cow level; the system determines when each cow should be milked using prediction models, udder refill patterns, behavior data, and robot load optimization. Overall, it is designed to significantly improve the milking process for farmers and provide them with

additional profits and higher milk yield. My part of the project won't be related to any production-level applications on the market for customers to use, but it is more centered on a research-oriented project aimed at finding ways to improve an aspect of the Beaver project by exploring the data and uncovering related patterns and insights.

Organizational chart:

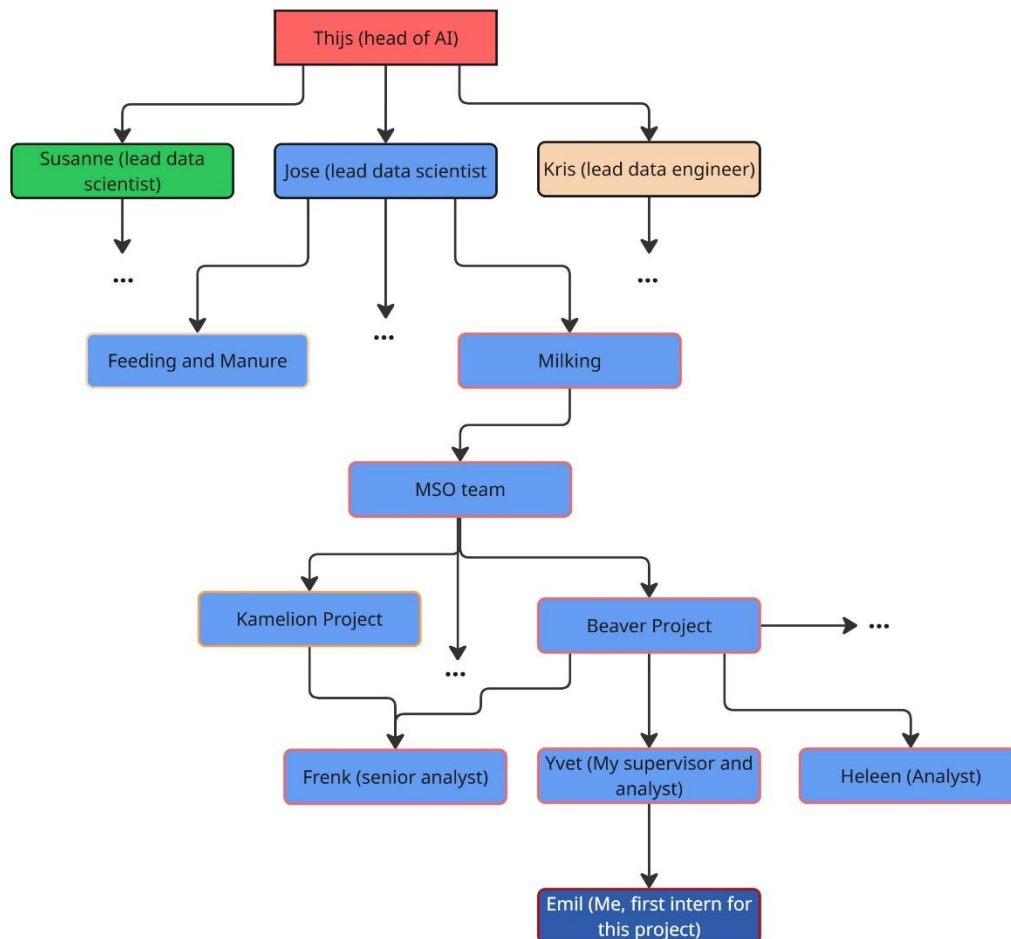


Fig.1. An organizational chart of the Data and AI team, making one for the whole company is outside of the scope and it also highlights my role in the company.

1.8 Work culture and environment:

The way I would describe the culture and environment at the workplace is quite relaxed and open, both within my immediate AI team and outside it, with whom I have had contact. Everyone was welcoming, and there were colleagues of all backgrounds, nationalities, and ages, and there were very few times in which I felt excluded or uncomfortable. I would describe the work culture, the way things work here, and the systems in place as quite free-flowing. Lely has a productive work environment without making it feel too robotic or work-heavy.

As this is a family-owned company still guided by its founders' original philosophy of practical problem-solving for farmers, the internal culture at Lely reflects the same directness. In my experience at Lely, I found that I could approach senior data engineers, managers, and higher-ups at the company without any formal escalation, and I was trusted to explore the AMS dataset independently rather than following a formal and rigid task list for my work.

This shaped how I worked at Lely, instead of awaiting detailed instructions on what to do and where to go, I took the initiative to test different methods I wanted to use and was curious about and then was able to present my findings not only to my immediate team but to other colleagues with different positions across the entire company, depending on what kind of feedback I was looking for. This mirrors the same autonomy-driven, pragmatic approach Lely expects from its own engineers.

There are regular monthly events and meetings with different subsets of the teams, where we all gather to discuss specific topics, catch up, brainstorm, etc. In my team specifically, which consisted of 5 people, we had weekly sprint meetings, bi-weekly team meetings, project sync calls, weekly standups with everyone, and I also had regular one-on-ones with my supervisor. Whenever I tried to set up meetings with other people in the AI department, they were always open to it and sometimes even made time. In the weekly sprint planning, each team member's tasks are reviewed for the coming week, with completion statuses for each.

After many years in operation as a family-run business, their values have barely changed, and the people who run the company still have the same mindset they had from the very beginning: Farmers come first. It's not always about profit; it has always been to make farmers' lives easier, and everything is done for the benefit of the farmers.

2. Ethical and Legal considerations

2.1 Databricks:

Over the course of my internship, I worked primarily in Databricks. Databricks is a cloud-based platform for data analytics and artificial intelligence. Databricks developed the 'data lakehouse' architecture, which combines elements of data warehouses and data lakes to manage both structured and unstructured data.

Lely leverages Databricks to revolutionize dairy farming by transforming vast amounts of data from robotic milking systems into actionable insights. Using a "data product" approach with the Databricks' lakehouse platform, Lely reduces data implementation time by up to 90%, optimizing cow health, farm efficiency, and data-driven innovation in agriculture. Lely, a leader in robotic milking, treats data as a product (data + metadata), which enables them to quickly share insights across the organization, accelerating the development of new applications.

All Lely employees are trying to comply with certain requirements, and regarding compliance, there is an agreement with Data Bricks on data confidentiality and data security. Data Bricks is also linked to a storage account, and that account and all the resources

connected with Data Bricks are being handled in terms of security by Azure on the cloud level. Data Bricks also maintains the lineage of data and the flow of the data, to see where it starts and ends.

Terraform is also used to manage and deploy resources, and most things of that nature are handled there. Terraform is an open-source infrastructure-as-code (IaC) tool that allows you to define, provision, and manage cloud and on-premises infrastructure using a declarative configuration language (HCL). With Terraform, it is used to granularly define permissions on the catalog of Databricks and all of its databases and tables.

2.2 Transparency & Explainability in Data:

For Lely, in the general sense, transparency and explainability are important. Not specifically in a very data-centric way, but in how the data appears to farmers, where it comes from, and how it is used. At Lely, using the large amounts of data for a specific farm, for example, there is always data being generated from it and data stored from that farm. So, we are able to see what kind of metrics it is producing and what metrics/farm strategies the farmers use, and if an anomaly comes up or something worth fixing/exploring, we, the programmers, can send optional 'advice' to the farmers to follow, for the most optimal outcomes. This means it is important for the data and the solutions we provide to be completely transparent and explainable to convince farmers to adopt these strategies and solutions, especially for stubborn farmers who aren't fond of new technologies.

2.3 Data Governance:

The core goal of data governance at Lely is to ensure data is managed in a way that generates value, not for its own sake, but tied directly to any business goals, process improvement, and risk reduction. To make this more manageable, all Lely data is divided into 13 data domains (e.g., customer, supplier, milking, feeding). Within those domains, data is further split into data products, which are sets of data actively used within the organization. The purpose of this division is accountability, ensuring there is always a clear owner for data quality and management. Without it, nobody could answer who is responsible for data quality or who to go to when issues arise.

The key roles for Data Governance include:

- **Data Domain Owner** - sits within the business itself, closest to the actual processes and the data those processes generate, responsible for maintaining data quality day-to-day
- **Data Product Owner** - operates at the tactical level, accountable for decisions about a specific data product
- **Business data steward** - sits within the business itself, closest to the actual processes and the data those processes generate, responsible for maintaining data quality day-to-day
- **Technical data steward** - the application expert, responsible for the storage, lineage, and technical management of data within a specific tool or platform (e.g. Databricks). One person can hold multiple roles simultaneously.

2.4 Compliance & Regulations:

Within Lely, there is an information classification policy that groups data and information into 4 categories: highly confidential, confidential, internal, and public, with documented definitions for each. But anything beyond that and compliance becomes less structured. For external regulation, GDPR is the primary standard the company follows, and a data privacy officer role has been introduced, but there is no formal or standardized procedure to ensure organization-wide compliance. People working with personal data are expected to consult either their domain owner or the privacy officer, but this does not always happen in practice. The EU AI Act is on the radar and being discussed internally, but, much like the GDPR, it does not yet have a formal compliance process in place at the company. An external consultant is currently drafting AI principles for the organization, though these are not yet finished or published.

2.5 Maturity Level & Ethical Considerations:

After assessing the company's legal and ethical considerations, it seems Lely is still in a very early phase of data governance maturity. Topics like fairness and bias in data and algorithms, such as unrepresentative training data or historical discrimination, are not currently being actively tested or monitored. In the future, as the company grows, the expectation is that, as the governance structure matures and the domain/product owner/steward roles become more established, these responsibilities will naturally be picked up at the appropriate levels. But for now, the focus is to get the foundational structure in place first.

2.6 Applying a framework and ethical assessment:

To critically assess how my work and Lely's broader data practices align with responsible AI principles, I apply the EU's assessment List for Trustworthy AI (ALTAI) (European Commission, 2020). ALTAI operationalizes the requirements set out in the EU AI Act; a regulation Lely is already tracking internally but has not yet formalized. Making this framework a fitting method for evaluating both the organization's current maturity and the decisions I made in line with the results I have found. Of the seven requirements that ALTAI has, I will focus on the 4 most relevant to Lely's context: transparency, accountability, human agency and oversight, diversity, non-discrimination and fairness, and privacy and data governance.

2.6.1 Human agency and oversight:

This requirement states that AI should support human autonomy and decision-making, not undermine it. Systems should be overseen by humans (European Commission, 2020). Lely's built-in system for sending out optional advice to farmers, rather than automatically altering the machine's behavior, directly reflects this principle. The farmers still retain full control over what settings and metrics they would want to change for their farm and dairy cows. The advice is just that: advice. They are not enforced and are completely optional.

This same principle also applies to my work and role on the MSO team. When I presented my findings to my team and other farmers, I did not present the cluster analysis as

prescriptive rules to be followed but as a starting point for discussion, leaving the final decision on which insights were actionable to domain experts with direct farming knowledge that I simply did not have. And beyond that, throughout the entire process, I always had the final say on my projects, and my supervision of the work was constant. Nothing was ever automated, nor could it be, as I approached everything with a human research mindset.

2.6.2 Privacy and data governance:

ALTAI's privacy and data governance requirement covers data quality, integrity, and access protocols, as well as respect for privacy across the full data lifecycle: collection, storage, use, and sharing (European Commission, 2020). As I wrote earlier in the paper, Lely does a good job meeting this requirement with the governance of Databricks' lakehouse, which includes confidentiality agreements. Access control is enforced through code and also maintains an information classification policy that sorts data into four tiers. However, as noted in section 2.4, the GDPR is the primary external regulation, but there is no standardized, organization-wide process to ensure compliance in practice. This creates a gap: the systems for protecting and classifying data are well-built, but the human process for ensuring those systems are used consistently remains informal.

In my own work, this was reflected in the kind of data I worked with and what I did with said data. There is almost no personal information revealed to me in the data I work with; everything is pseudonymized before it even reaches my colleagues or me. Things like farms and animal names had a code rather than a name. And the privacy of work and data is strictly enforced in the company. Anything regarding this highly confidential information cannot, under any circumstances, be revealed to anyone outside the team you are working with or to people under contract with information protection guidelines. It is so protected that even I couldn't reveal the names of my projects or details of what I did during my comeback presentation. And all data from farms is classified as the Farmers' data they own, not Lely's. The fact that data is protected by an architecture that operates before it reaches me illustrates how privacy at Lely is currently handled: enforced by design in the pipeline, rather than through a governance process I was ever required to actively engage with.

2.6.3 Transparency:

This requirement holds that AI systems should be traceable, explainable, and clearly communicated to those affected by their outputs (European Commission, 2020). Lely partially complies with this requirement. They do have optional advice generated by the AMS machines, which explains what the new advice is, what it means, and how the decision to provide it was made. Databricks' built-in lineage tracking also supports technical traceability, allowing engineers to see where specific datasets or tables originated, how they were transformed, who owns them, and who the most frequent users are. However, this transparency is stronger externally than internally. While farmers are given simple explanations of new decisions, there is no equivalent plain-language explanation or documentation for internal non-technical stakeholders to understand how a specific model was produced. There is documentation for many of the models being made and trained, but it

is usually reserved for those who can understand every technical aspect mentioned in the document.

In my project, I addressed this by explaining every decision I made and the changes I made, detailing everything in the notebooks I used, with each step of my process including an explanation of what I did. And of course, I did the same in my regular meeting with my team and other colleagues. I also made handover documents for my supervisor and anyone else who would be taking on my project in the future, with detailed explanations included.

2.6.4 Diversity, non-discrimination and fairness:

ALTAI requires AI systems that avoid unfair bias and are evaluated for their impact across different groups affected by them (European Commission, 2020). As stated in section 2.5, Lely does not yet have active testing for bias and fairness in its models or data, and this gap was evident to me during my analyses. Specifically, when I conducted my cluster analysis, on many occasions for certain clusters the underlying data for that cluster was less reliable than for the other ones. Rather than presenting the recommendations from those less reliable clusters, I presented them as a limitation of the data to the colleagues to whom I was presenting these findings. This was a deliberate informal application of this requirement; had I presented them uniformly with the same weights as the other clusters, farmers whose data fell into lower-quality clusters could have received recommendations based on less trustworthy evidence than farmers in other clusters, therefore producing an uneven quality of service without any technical bias being present.

2.6.5 Accountability:

ALTAI's accountability requirement states that mechanisms of responsibility, auditability and redress are to be clear and explained when an AI system's outcomes cause harm (European Commission, 2020). Lely's data governance structure divides data into 13 domains, each with a designated Data Domain Owner, Data Product Owner and Business/Technical Data Stewards. These facts give this requirement a reasonably strong foundation, since the responsibility for data manipulation, protection, and ownership is assigned rather than left ambiguous. However, this accountability structure operates mainly at the data level, not at a model level. As I mentioned in section 2.4, there is no standard compliance procedure for third-party frameworks such as the GDPR or the EU AI Act, and adherence currently depends on individuals contacting a domain owner or privacy officer directly, which in practice does not always happen. This means that Lely is accountable for who owns the data, but has a much weaker structure for who is accountable when a model's recommendation is wrong or causes harm to anyone. And the same goes for my models' recommendations, although my work won't ever be applied to real, active products that clients use.

3. Internship Task: Explanation and Approach

In dairy farming, the dry-off period refers to the phase in a dairy cow's reproductive cycle during which milking is deliberately ceased, allowing the cow's mammary gland to rest and regenerate before the next lactation. This transition period, typically lasting around 6 to 8 weeks prior to calving, is critical for udder health, milk production in the subsequent lactation, and overall cow welfare. Effective dry-off management, including the timing of cessation, nutritional adjustments, and the use of dry cow therapy, plays a significant role in determining the herd's productivity and health outcomes. As dairy operations scale and data availability grows, accurately predicting and managing milk production traits around the dry-off period has become an increasingly important area of research.

My internship focuses heavily on the dry-off period of the cow, and, in general, I am confined to the end-lactation stage. My task was to gain insights and findings into this period of the cow using the vast datasets collected from the thousands of farms that use Lely robots for milking and relate these findings to outcome variables such as the milk yield of the cows or cow health, and give strategies about moving forward with the dry-off process. The project I am involved with does not immediately concern improving the Beaver project, such as improving advice or automating advice, or any Lely products or services currently on the market for customers to use. It is a more research-oriented project aimed at studying this specific subfield of the market and the Lely product, and can be used to create a prototype directly from my research and testing.

Relating all this domain knowledge to my own skills and background, we decided the way to gain the most insights was to use clustering. Specifically, I chose K-Means as the algorithm to cluster different settings used by farms at the end of lactation. Each farm and group of cows had a different combination of these settings, so the goal was to use the settings each farmer had set and perform a cluster analysis with them in mind. Then I would get back the most-used combination of settings and strategies each farmer uses and relate it to certain effects, such as daily milk yield in the animals during the end-lactation period. Moreover, with that, I could generate cluster profiles by combining those settings to gain insight into why a farm would use a specific strategy for its cows.

4. Integration of research and theory

The use of Machine Learning (ML) in the broad field of Agri-tech is nothing new to the industry, for both agricultural companies and in Data Science and AI. Focusing more specifically on the application of machine learning to dairy farming and Automated Milking Systems (AMS). All of it spans a range of objectives, from disease detection and herd characterization to milk yield prediction and farm-level classification. This review synthesizes findings from relevant studies, examining the methods employed, results achieved, and gaps that remain in the literature.

A broad review by Ozella et al. (2023) of 60 other studies that used ML algorithms and applications in AMS revealed an imbalance in the field's research focus. The existing work focused mostly on disease detection, with milk production receiving little attention, the topic I will focus on. This highlights a clear gap in research and a lack of attention to milk production in the field of ML and dairy farming. This is surprising given that AMS systems continuously generate rich production data that could meaningfully support farm management decisions.

Many of the studies reviewed explored AMS data for herd characterization and to monitor cows' behavior to uncover and develop unique patterns over time. Ozella et al. (2023) reference the work of Adamczyk et al., in which the physical activity of dairy cows milked in the voluntary milking system, recorded by neck-mounted tags, was analyzed using ML methods. They performed a cluster analysis, classifying cows' physical activity using Ward's hierarchical agglomeration method and Kohonen self-organizing maps. Physical activity showed low variability within individual months. However, across months, the clusters obtained by Ward's method highlighted different groups depending on daylight length, temperature, and relative humidity. In that study, Kohonen networks were used only to verify Ward's method and to test the similarity between clusters obtained with both methods.

The same review also highlights K-means clustering applied to herd segmentation using five parameters: daily milkings, parity, activity, milking regularity, and body weight, demonstrating the suitability of AMS data for automated, management-relevant grouping.

At the farm level, Žgajnar (2024) applied a combined hierarchical and K-means approach to classify Croatian dairy farms into representative archetypes for policy analysis. Their results showed that variable selection significantly affected cluster interpretability: a reduced two-variable model produced more logically ordered clusters than the full four-variable version, ranking farms cleanly by herd size and milk output.

Atıl and Akıllı (2016) directly compared K-means and artificial neural networks (ANNs) for classifying 10,000 dairy cattle by milk production traits. The ANN used a multilayer perceptron trained with three backpropagation algorithms, with Bayesian regularisation performing best. The ANN achieved a peak R^2 of 97.8%, compared to 94.9% for K-means, though performance varied across datasets, dropping to 72.3% on one, indicating sensitivity to data characteristics. Overall, the ANN demonstrated a better fit, suggesting neural networks are a viable alternative to traditional clustering for animal classification.

Ji et al. (2022) developed an XGBoost-based forecasting framework using 5 years of data from 80 cows on a robotic dairy farm, predicting daily milk yield, milking frequency, protein content, and fat content up to 28 days in advance. Validated across 42 cross-validation folds, results were strong: protein content reached ~98% accuracy, fat content ~95%, and daily milk yield maintained a mean above 90%. Feature importance analysis showed that cow-level features were considerably more predictive than climate features, and that cow age outweighed lactation stage, which is attributed to the shortened productive lifespan of modern cows, creating pronounced age-related production differences. A key limitation was systematic bias at extremes, with the model overestimating low values and underestimating high ones, a common pattern in large-dataset regression models.

4.1 Synthesis

Across the literature, K-means combined with Ward's method emerges as a consistent and versatile tool for both cow- and farm-level segmentation. ANN approaches outperform K-means in direct comparisons, though not uniformly. For production prediction, XGBoost shows strong performance when cow-level features are prioritized. The central gap identified by Ozella et al. (2023) and reflected across all reviewed studies is the underrepresentation of production-focused ML research relative to disease detection. Future work could benefit from integrating clustering and prediction within a unified framework, using herd segmentation outputs as contextual features for individual-level forecasting.

4.2 Methodological literature

The studies I discussed above establish clustering as common in dairy analytics, but the literature says little about the specific methods underlying my own analysis (e.g., PCA, autoencoders).

Brotzman et al. (2015) applied Principal Component Analysis (PCA) to reduce 22 herd-level variables to 16 before clustering 557 Upper Midwest dairy herds, a methodologically close approach to my own reduction of AMS variables before clustering end-lactation settings. More recent work extends this to sensor data specifically: Riaz et al. (2025) found that PCA-based dimension reduction improved downstream model performance on high-dimensional dairy cattle accelerometer data compared to using raw features. Similarly, Matzhold et al. (2024) identified PCA and UMAP as core techniques for addressing the increasing dimensionality of AMS and precision livestock sensor data prior to clustering. This mirrors my own challenge of reducing a large AMS feature set to something clusterable.

As a non-linear alternative to PCA, Lee et al. (2022) used PCA and a deep autoencoder to extract features from survey data, clustered with k-means. They validated results using the Calinski-Harabasz index, silhouette coefficient, and within-cluster sum of squares. Ji et al. (2024) similarly combined a Contractive Autoencoder with K-means and Silhouette Score, which outperformed alternative reduction methods in an industrial sensor context. This suggests that an autoencoder could have captured non-linear relationships in AMS variables (e.g., between udder health, milking frequency, and yield) that linear PCA cannot, which is a limitation worth acknowledging in my own methodology.

During my work, I used the silhouette score as an accuracy metric. This metric measures how well data points cluster with similar points versus separate from other clusters, ranging from -1 (poor) to 1 (belongs in cluster) (Rousseeuw, 1987), and appears consistently across domains as the standard cluster validation method, reinforcing that my own use of it follows established practice rather than being an isolated choice. Appears consistently across Lee et al. (2022) and Ji et al. (2024).

Going down the same path, it must be mentioned that the Elbow method was also used to calculate the appropriate number of clusters to use. The elbow method identifies the points of diminishing returns in the within-cluster sum of squares as the cluster count increases (Thorndike, 1953), a commonly used method for K-means in this domain.

5. Methodology

5.1 Exploratory Data Analysis

Before I started the cluster analysis, which would lead to the results and final deliverable for my task, I had to conduct some data analysis. I asked questions that were in and around my main task to first gain insights and context, and to better understand the kind of data I was working with so I could move on to the main part of my work. This step of the process was also used by me to gain domain-specific knowledge, to get a real grasp of the data and how to use it before I got into clustering. The first step is always to understand the data.

The main language used was SQL for all pre- and post-analysis, all done within Databricks. The methodology for the analyses was first to formulate questions about the main topic that I sought to answer with the data, which could later help with the main task.

It all stemmed from a big query given to me by the team, in which the features/settings I used for clustering were all calculated there. After exploring the data and preparing the query for the main task, I saved it as a table for easier access later, after which I started on the main task.

5.2 ML tools

The cluster analysis was conducted using four key settings that farmers configure for their cows during the end-lactation phase, derived from the milk access table (MAS). These settings were selected as input variables as they directly reflect the dry-off strategy applied by each farm:

- **Days until dry-off:** the point at which a cow transitions from mid to end lactation in the milk access table
- **Minimum number of milkings in end lactation:** the lower bound of milking frequency permitted during the end lactation phase
- **Maximum number of milkings in end lactation:** the upper bound of milking frequency permitted during the end lactation phase

- **Optimum expected yield in end lactation:** the target milk yield set for a cow during the end lactation phase

Then, to deduce the number of clusters to use, I tried the elbow method (Thorndike, 1953), which measures the squared distance between data points and their assigned cluster centroids. The most common number of clusters I kept getting was between $K = 4$ and 5. Whenever I tried using more clusters, the latter end of them, mostly the clusters 6 and upwards, would always overfit, and then it would underfit with too few clusters.

To be able to graph all the clusters as points on a scatter plot, and due to the nature of this case being more than 3 dimensions (4 settings were involved, more than 3), I had to conduct a Principal Component Analysis (PCA) in order to project the results on a 2D plane. Then I combined the settings into Principal Components (PCs), based on how much they affect the results, so the most influential settings were grouped into PC1 and the secondary ones into PC2.

5.3 Quality and Reliability

Throughout the process of carrying out my internship tasks and project, while I was writing code logic, exploring new ideas and insights that revolved the main project, any model that I decided to develop, to ensure real validity and quality to the solutions I provided or methods I chose to work with I always made sure to check my work over with my supervisor and the rest of my team to make sure everything is up to standard and that I was not receiving false information that would hurt the analyses.

Since I came into this with little to no knowledge regarding the dairy and farming industry, making sure to check everything over with them was of utmost importance to ensure no metrics were subject to hallucinating values and that the results I was getting were realistic and always made sense.

In addition to consulting with my team and colleagues, I consistently conducted data validation checks to ensure dataset accuracy.

5.4 Adding a new feature

After discussing and analyzing the previous results, I decided to add a new layer to my analysis. That came in the form of making a new feature or setting for the clustering analysis. This was done to improve clustering; the main aim was to capture farm-level grouping behavior among late-pregnancy cows. Since my whole project focused on late-stage cows, adding this extra layer would enrich the analysis of late-stage lactating cows. The goal of this feature is to incorporate biological and management-based context into the clustering, beyond purely configuration-based MAS parameters.

While the current variables describe the system configuration and touch on important features used in the different farming strategies currently in use, they do not capture how farms organize cows into groups, whether farms use late-pregnancy grouping strategies, and how management aligns biological state with MAS rules. To address this gap, this new feature was added.

The new setting was made with a combination of two separate metrics:

- `animal_days_pregnant_from` ≥ 200
- `animal_days_pregnant_to` ≥ 200

These features indicate whether the farm uses late-pregnancy rules, defined as the two metrics above. These groups correspond to cows approaching the dry-off phase, where milk access restrictions are typically most relevant. The ≥ 200 means I filtered for cows that are at least 200 days into the late-pregnancy rule.

I implemented it in the main SQL query used for the cluster analysis as a feature by transforming it into a percentage. It starts by checking the ≥ 200 rule in either case and assigning a 1 or a 0 depending on whether there is any record of the rule's usage for any cow. Then that binary score is averaged across the entire cluster, multiplied by 100 to get the percentage. And with that, the new feature is obtained.

5.5 Extra considerations

5.5.1 Important flags

For the main SQL query used to perform the cluster analysis, several important considerations had to be kept in mind to ensure the validity of the results. That comes in the form of flags that some of the tables I used include. Those are:

- **is_current_version flag:** This flag checks if the settings and metrics of the cow that is being queried and studied in the current moment are the valid ones to use and the active true settings being used by the farmers for the cows for the current moment. This is important because Lely keeps a record of historic settings used by all the cows, as these animals go through many stages of lactation during pregnancy. And it is always important to ensure I am using the current settings; otherwise, none of the results can be used.
- **group_deleted flag:** In every farm, there are different sets of groups for different stages of cows, each with their individual name and ID. Oftentimes, when one of those groups has finished that stage, and there are no other groups to replace it, the group is deleted. So, in this case, it is also important to keep only existing groups in the analysis.
- **active_flag:** This, much like the other two, is important to have enabled to make sure the analysis only takes in cows that are currently active and being milked regularly. There is also a common table expression (CTE) at the start to filter only for animals that have been active in the last 40 days and are in the late stage of lactation.

5.5.2 Thresholding

As noted above, additional filtering measures were considered in the analysis. Instead of checking binary flags, we used a value threshold as an additional analysis strategy. After checking the max threshold for each of my features in the Horizon application, the main application and dashboard that farmers use to monitor the animals, provided by Lely, I could gather what the max setting that can be used within the application is. This must be done

because, a lot of times, farmers somehow input settings that are very irregular and impossible. Whether that is a fault of the system or of the farmer not knowing what they are inputting or how they are doing it, it is important to set a cutoff limit for realistic values to use within the analysis. The cutoff values include:

- **Days until dry-off ≤ 30**
- **Minimum number of milkings in end lactation ≤ 21.6**
- **Maximum number of milkings in end lactation ≤ 21.6**
- **Optimum expected yield in end lactation ≤ 20**

5.6 Using ANN's

Earlier in section 4, Atıl and Akıllı (2016) demonstrated that ANN's outperformed K-means for classifying and clustering dairy cows. I decided to try that approach as well but with unsupervised machine learning algorithms instead. However, their approach involved a multilayer perceptron trained on labeled data. Since my own data did not have ground-truth labels, I decided to choose an autoencoder instead. This gives the cluster analysis a better learning structure from the unlabeled data. Since an autoencoder only needs the raw data itself as input, it is well-suited for clustering tasks where ground-truth labels do not exist.

6. Results

6.1 First general cluster analysis

The final table obtained from the clustering is the following:

Cluster	Max milk hr	Min milk hr	Milk yield kg	Dry off days
0	7.89	10.64	9.06	33.48
1	17.13	26.10	13.02	15.93
2	7.88	10.32	8.55	14.46
3	9.64	12.78	11.07	15.37
4	46.04	56.60	10.77	6.28

Fig.2. General cluster analysis table revealing all of the base settings

The distribution of farms for each cluster is the following:

Cluster	Nr of farms
0	5850
1	1186
2	12377
3	10258
4	180

Fig.3. Table showing the distributions of farms per cluster

As can be seen, the bulk of the farms are within cluster numbers 2 and 3, which happen to be the most standard combination of settings farmers seem to use. Then the more irregular settings are more concentrated in the other clusters, notably in the last one, where there are only a very small number of farms located within it. Going back to the table, that last cluster

has by far the most irregular settings if you compare them to the other clusters, meaning the farmers that are in those clusters have quite the irregular combination of settings and strategies for their animals.

The distribution of weights for PC1 and PC2 loadings is the following:

Feature	PC1	PC2
Max milk hr	0.65	0.06
Min milk hr	0.64	0.11
Milk yield kg	0.34	0.18
Dry off days	-0.17	0.97

Fig.4. Table PCA loadings and distributions

From the table, we can observe that PC1 has strong positive loadings for max and min milkings hrs, so when a farm is high on PC1, it usually means they are quite loose with their milking hours, having a larger window for milkings. If it is low, then it means the milking time window is quite tight.

For PC2, it has positive loadings for milk yield and dryoff days. So, when a farm has high PC2, it means the milk yield is high, and the dryoff window is long; the opposite holds if PC2 values are low.

Here are the projected PCs on a scatterplot. Each point represents a farm from the dataset:

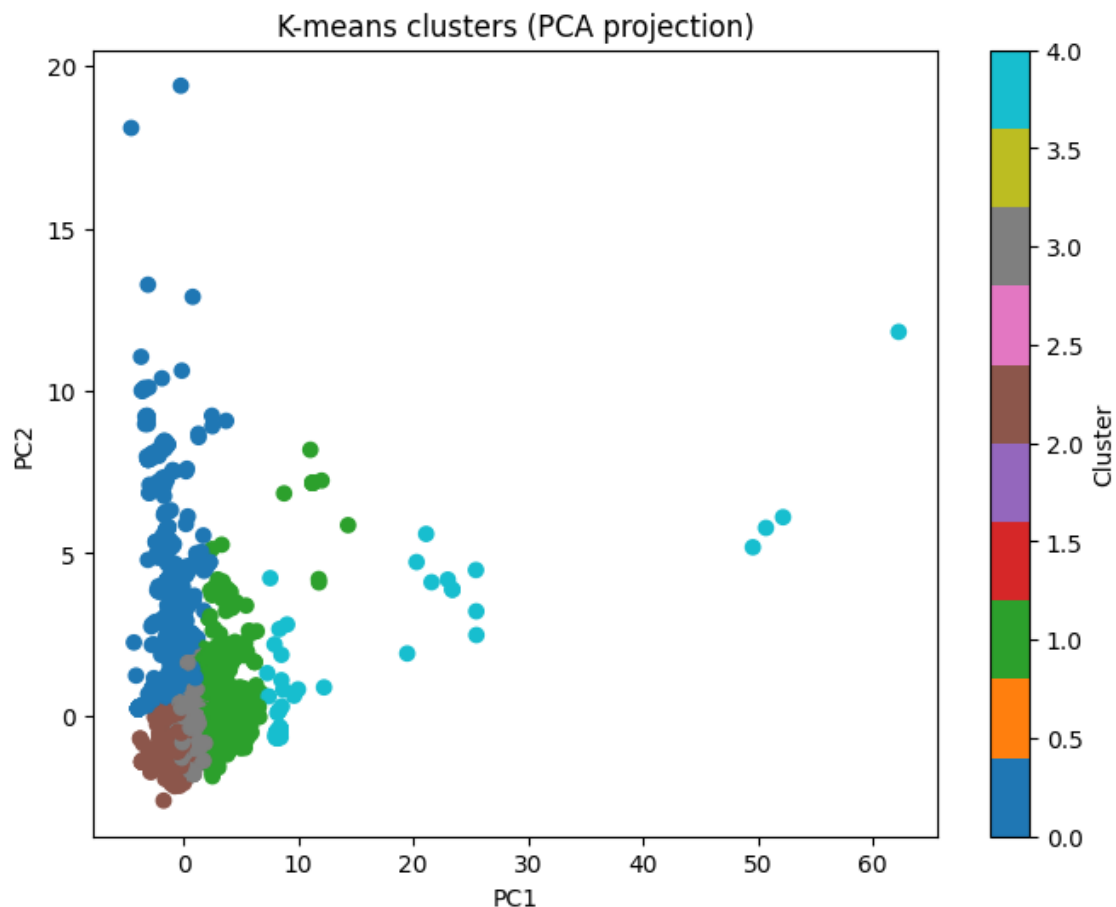


Fig 5. Scatterplot of PC1 and PC2 (PCA projection)

Observing the scatterplot, the more unusual combinations of clusters, such as number 4 or 0, are much more scattered and tend to fall at more extreme values, as can be seen in both the scatterplot and the general cluster table.

As a test to measure the accuracy of the clustering, a silhouette score was calculated to quantify how close each point in the cluster was to the other points in the cluster. The silhouette score here was 0.3181. The closer it is to 1, the better, and the closer it is to -1, the worse.

The derived metrics and results from this cluster analysis for the amount of milk yield kg were the following:

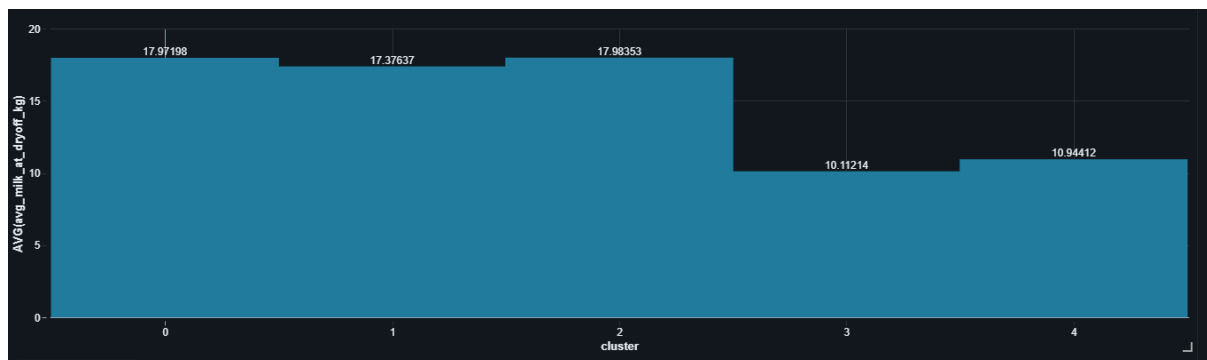


Fig.6. Bar chart showing milk yield kg at day of dry-off per cluster

As can be seen, the more populous clusters had higher average milk yield. This highlights the limitations of the strategies used for clusters 3 and 4 and points out areas for improvement.

6.2 Other methods for analyses

6.2.1 Using the new feature

Earlier in the document, I talked about doing some feature engineering and creating a new feature for the cluster analysis called **has_pregnant_group**. The silhouette score achieved for this iteration was 0.4797. A slightly lower score, but by a very negligible amount.

The extra feature's results and the numbers it garnered might explain these outcomes.

Cluster	has_pregnant_group (%)
0	0.039
1	0.057
2	0.074
3	4.907
4	0.028

Fig.7. Table indicating the avg usage of the has_pregnant_group rule in percentage value

After looking at the results, only a very small number of farms use this strategy, as it is not common. Aside from cluster number 4, which, compared to the rest of the clusters, has a very high amount of this happening. This helps determine the true strategies and methods for each cluster, as the 4th one provides a clearer picture of the weights used for the clustering itself.

6.2.2 Using ANN's

When I switched from traditional clustering to implementing an autoencoder into the cluster analysis, I went through many iterations of different versions of the analysis. The most important ones will be highlighted. Because of the nature of neural network weight initialization and training, results can vary between runs even with the same hyperparameters. To address that, the model was trained and evaluated using 5 randomly selected seeds, with silhouette scores reported for each run. The mean and standard deviation across those 5 runs are calculated, rather than a single individual score, yielding a more statistically robust interpretation of model performance.

6.2.2.1 Base query, no extra feature and thresholding applied

In this iteration, I used only the base query, which includes 4 features, and applied no thresholding. The number of clusters here is 5, running over 150 epochs and a bottleneck of 16.

The mean silhouette score ended up being 0.4436. The distribution of farms to clusters ended up being:

Cluster	Nr of farms
0	18706
1	4232
2	325
3	3
4	3251

Fig.8. Table indicating the distribution of farms per cluster

This distribution of farms is quite unfavorable, as having one cluster populated by only 3 farms is not acceptable for this analysis. But even with that, the silhouette score shows a big improvement over the regular clustering analysis. This might be saying something.

6.2.2.2 Extra feature and Thresholding applied

For this iteration, I used the base query with the new feature applied and applied thresholding. The remaining values are unchanged, including the cluster number, which is 5.

The mean silhouette score here is 0.3861. And the distribution is the following:

Cluster	Nr of farms
0	16131
1	658
2	1423
3	74
4	7653

Fig.9. Table indicating the distribution of farms per cluster

As observed, the silhouette score has worsened, but the distribution of the farms has improved slightly here. Still, this distribution would still have to be classified as unfavorable, as the distribution is still lacking for this analysis.

6.2.2.3 Only extra feature applied

In this iteration, I only included the extra feature in the analysis. The rest was the same.

The silhouette score was 0.4859, and the distribution was the following:

Cluster	Nr of farms
0	7227
1	158
2	18615
3	456
4	26

Fig.10. Table indicating the distribution of farms per cluster

This time, the score improved again, but the distribution was still lacking.

6.2.2.4 Only Thresholding applied

For this iteration, the only change was the removal of the extra feature. Everything else stayed the same.

The mean silhouette score in this case was 0.3468. Moreover, the distribution ended up being:

Cluster	Nr of farms
0	15473
1	441
2	4158
3	4510
4	1361

Fig.11. Table indicating the distribution of farms per cluster

This time the results are more interesting. The silhouette score for this iteration was lower than all other scores using ANNs, but in terms of distribution, this is the best we have seen so far. There is some discrepancy here, and a trend of something specific happening.

6.2.2.5 Changing cluster size with no extra feature or thresholding applied

In the next iterations, I will choose a cluster size of 4 to see how the overall distribution changes and how the values for each cluster shift to the new size. As stated, this one has only 4 clusters and no extra features or thresholding applied; everything else stays the same.

The mean silhouette score was 0.5110. And the distribution was:

Cluster	Nr of farms
0	328
1	5195
2	11
3	20989

Fig.12. Table indicating the distribution of farms per cluster

The silhouette score was not affected at all; it actually improved as I reduced the number of clusters, but the distribution was once again not very good, with far too few farms ending up in cluster 2.

6.2.2.6 Changing cluster size with extra feature and thresholding applied

For this next one, the cluster number stays at 4, with the extra feature and thresholding applied to it.

The silhouette score for this iteration ended up being 0.5012, and the distribution of the farms was the following:

Cluster	Nr of farms
0	23570
1	655
2	1636
3	78

Fig.13. Table indicating the distribution of farms per cluster

Once again, the silhouette score was very strong when the number of clusters was reduced, achieving the second-highest mean score across the entire analysis. However, as in the last iteration, the distribution remains quite poor. Both results have different distributional structures, but they end up being similar when you weight the farms in each cluster comparatively.

6.2.2.7 Changing cluster size with only thresholding applied

In this iteration, only the thresholding is applied; the rest continues being the same.

The silhouette score ended up being 0.3751, and the distribution was the following:

Cluster	Nr of farms
0	15488
1	1751
2	4190
3	4514

Fig.14. Table indicating the distribution of farms per cluster

Once again, after we removed the extra feature and only had the thresholding applied in the results, we see the same phenomenon happening again. The silhouette score decreased dramatically compared to the previous two iterations, but when observing the distribution, it appears better distributed than the other two, making it look more suitable for analysis.

6.4 Discussion

After testing the autoencoder with the K-Means model many times over, interchanging parameters throughout, the results I ended up with were not exactly what I had hoped for. However, they still yielded interesting insights worth paying attention to.

A general pattern emerged after running all these iterations. For this, I will use the different conditions under which the iterations were run as conditional metrics for the analysis. If you look at it closely, this is what can be seen (for K=4):

Situation nr	Nothing applied	Extra feature applied	Thresholding applied	Result (silhouette score)
1	True	False	False	~0.51
2	False	True	True	~0.50
3	False	True	False	~0.50
4	False	False	True	~0.37

Fig.15. Iteration matrix testing different conditions

The key difference between the situations is that the last one has a lower average score than the others; the majority seem to share roughly the same average score. The negative condition for these results is in two forms: either nothing was applied (no extra features or thresholding were applied during the iteration), or an extra feature was applied. Whenever nothing was applied was True, the score always came out to ~0.51, and whenever the extra feature was True, the score came out to ~0.50, regardless of whether the thresholding was applied or not; in every situation, it would always overpower it. Then, as soon as the negative conditions were False and only Thresholding was True, the score always lowered to around ~0.37.

This means that the other two conditions, the negative conditions, carry more weight in calculating each cluster and its values. The reason is that whenever thresholding is applied, it excludes the extreme values I encountered many times from the cluster analysis. The thresholding only considers the most typical values observed across all farms. Even if the thresholds are set for the absolute max values for each setting a farmer can put for their milk access settings, which raises a different concern altogether about threshold limitations for the Lely robots themselves, those values are still quite normal compared to the overblown ones that could be found before. But the thresholding re-organizes all the values and consequently makes it harder to separate each farm into different clusters, since the values are not as extreme here, which is why the silhouette score is lower.

The reason the negative conditions take over and the scores tend to be higher is that those extreme values are included in the analysis. It is much easier for the model to classify farms into clusters when extreme values are present, as those values almost always constitute the overwhelming minority and, in many cases, end up forming a separate cluster or pair of clusters, with those extreme values defining it. Therefore, it is easier to group the extreme values than the more typical ones, resulting in a better score. That works the same way for the extra-feature condition. Since almost no farms use it, the cluster that populates it tends to be a huge minority. And in the results, the case tends to be that only one of the clusters is using said feature, leading to a better silhouette score. This is also supported by the scatter plot (PCA projection) made with only the thresholding condition applied:

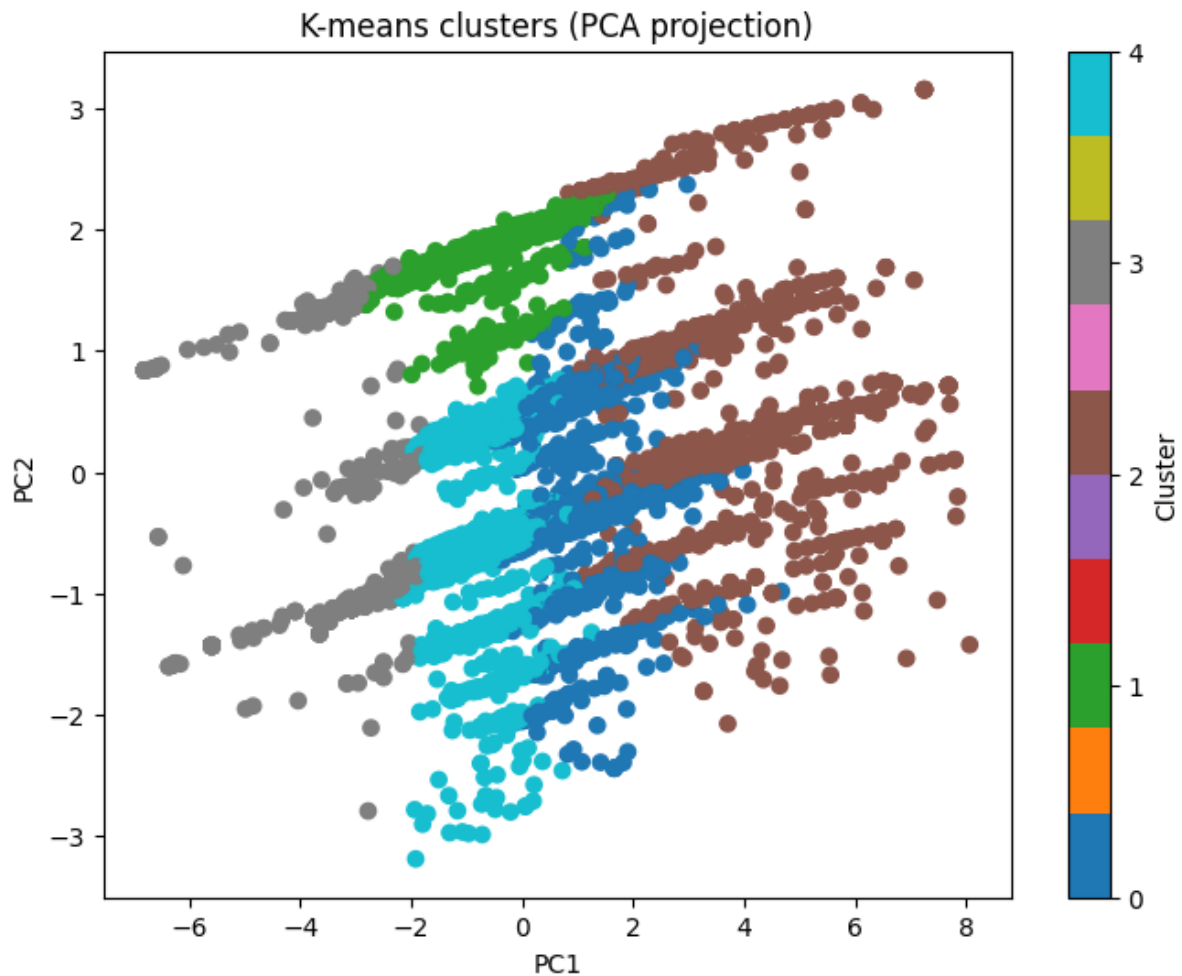


Fig.16. Scatterplot of PC1 and PC2 (PCA projection) with thresholding applied this time.

The clusters are broadly separated along PC1 (horizontal axis): blue is mostly on the left, green on the right, and grey in the top-middle. This suggests the clustering has found some real structure. The diagonal stripe pattern suggests the data form diagonal bands rather than round blobs, indicating that the features have strong linear correlations with each other. This is common in MAS settings; farms that allow more milkings also tend to set higher yield thresholds, etc. PCA is picking up on this linear structure.

The main differentiator between the two graphs, as explained by the negative conditions, is that 1 or 2 clusters are clear outliers with extreme values. In the first scatter plot featured, the extreme separation of the grey cluster is real information; it shows that cluster 3 is different from everything else, not just slightly offset.

6.4.1 Model limitations with past work

Trapanese et al. (2025) had a case similar to this one; they used routine test-day recording data (regularly collected production and physiological metrics) from three Italian Mediterranean buffalo herds, applying K-means and hierarchical clustering to determine whether data-driven groupings could support herd management decisions. Structurally, it was

similar to what was done here with the farm settings, but applied to individual buffalo rather than farm-level configurations.

They ran the analysis two ways: once on all three herds combined, and once on each herd separately. They found that K-means consistently outperformed hierarchical clustering across all datasets, shown through average silhouette scores of 0.17-0.18 for K-means versus 0.10-0.12 for hierarchical clustering, along with more favorable Davies-Bouldin Index and Calinski-Harabasz Index values for K-means. K-means found two clusters in the combined dataset and in two of the three individual herds, while the remaining herd split into three clusters.

When two clusters formed, days in milk and milk yield were the main variables separating the groups; when three clusters emerged, K-means also identified a subgroup that differed in age and lactation stage. These cluster differences were statistically confirmed via ANOVA across most variables studied.

Comparing the results from this study to my own, it can be gathered that my silhouette score averages higher across most of the tested variables. The lower scores in the mentioned study are likely a reflection of the buffalo test-day data, which may have less natural separation than the more configuration-driven MAS settings I clustered. This also strengthens my justification in using K-means over a method like hierarchical clustering due to the difference in scores. It also confirms that K-means over hierarchical clustering is a closely related domain.

A different study conducted by Michelena et al. (2025) compares three unsupervised anomaly detection methods to each other. The methods were: DBSCAN, Local Outlier Factor (LOF), and Isolation Forest. It was used to detect estrus (heat) in dairy cows, based on daily behavioral data from smart collars on 22 cows across two Spanish farms. They tested two model topologies: one model per individual cow, and one aggregated model per farm herd.

The results showed that individual cow models performed well with F1 scores above 90%. Hard-level models performed poorly, with F1 scores around 75% (farm 1) and 62% (farm 2). They found that no individual technique dominated; depending on the cow, each method won in different cases. Using t-SNE visualization, they showed cows have genuinely distinct, individualized behavioral patterns, which is why a single herd-wide model underperformed.

Linking this to my case and why K-means might have been better suited for my project: in the study, DBSCAN's role was anomaly/noise detection, not group clustering. This means that DBSCAN's strength lies in flagging outliers rather than forcing them into clusters. This was not tested in my project, but it is suspected that if DBSCAN were used in place of K-means, it could have played the role it did for the extreme/outlier farms in the data I used. Beyond that, no study was found on a direct comparison between clustering models with the exact same context/content I worked with.

6.4.3 Silhouette score limitation

Although the silhouette score is usually a good metric for evaluating clustering and other types of algorithms, it, too, has its limitations. The pattern pointed out in the earlier section reflects a known limitation of the silhouette score: it is sensitive to cluster-size imbalance and can be inflated by small, well-separated outlier clusters rather than by genuine overall cluster quality (Rousseeuw, 1987). My own results might be an indicator and reflection of this limitation. The iterations with the highest scores also had the most imbalanced distributions, meaning the silhouette score alone might have been an incomplete indicator of practical clustering strength in this case.

6.4.4 Impact of work

These findings support the Beaver project's objective of improving the timing of milk access to increase farmer profit and milk yield. The clusters and the PCA revealed that the farms cluster distinctly along two main dimensions: how loosely or tightly farms configure their milking-hour windows (captured by PC1, influenced mostly by mix/max milkings settings) and the combination of milk yield and dry-off duration (captured by PC2). The final cluster profiles show a clear split in outcomes; clusters 0, 1 and 2 achieved a milk yield of roughly 18 kg on the day of dry-off, while clusters 3 and 4 achieved only around 10kg, a substantial gap that can be tied back to differences in configured settings more so than random variation between farms.

This presents the MSO team with an evidence-based starting point; instead of treating all farm configurations as equally viable, the Beaver project's prediction models and the Horizon application's advice generation logic could specifically flag farms whose settings resemble the lower performing cluster (3 and 4) and shift more of the needed guidance to encourage changes and adjustments towards the configuration patterns that are associated with clusters 0-2. This then starts to move the analysis from a purely explanatory grouping exercise into a more practical decision support system (although like mentioned before the project is still only research based and not made for production), identifying not just where farm strategies might differ, but which specific combinations of settings are associated with a better outcome for milk production, giving Lely's advice system a more data driven basis for prioritization of which farms to target and what kind of advices and changes to recommend for said farms.

6.5 Interpretation

After seeing all of these results and going over the explanations for them, it would not be too far-fetched to conclude that the new extra feature, the `has_pregnant_group` feature, is obsolete in this analysis, not contributing much to it, and, more so, hindering it in the long run. And that would be a fair assessment of the feature, but there is something underlying in these results that we can take back with us.

The fact that the values are so extreme is both a hindrance and a help to the analysis. After applying the extra feature condition, or not applying either of the conditions, what is revealed are the combination of settings that should not be used and help distinguish between good

strategies and the bad ones; therefore, those farms that are in the more extreme clusters have an easy path for improvement and know exactly what needs to be better.

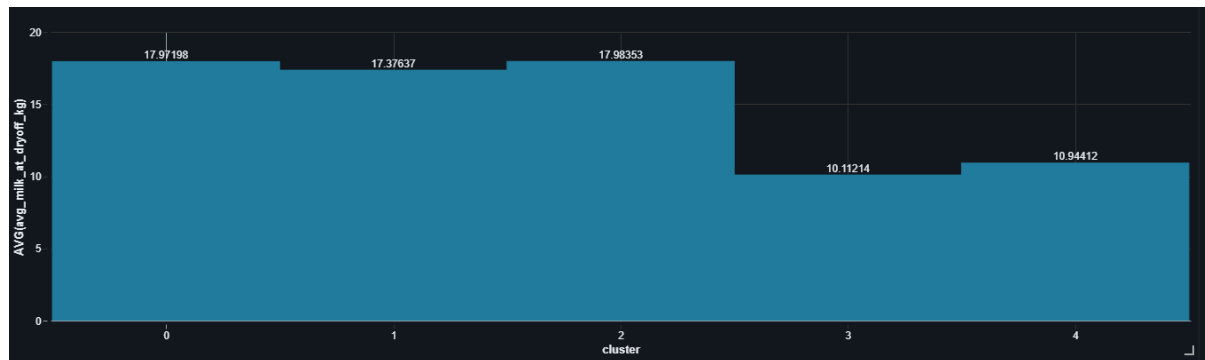


Fig.17. Bar chart showing milk yield kg at day of dry-off per cluster

Going back to this graph, which shows the average milk yield on the day of dry-off for each cluster. The first three clusters yield a high 18kg milk yield, while the last two show only 10kg milk yield, which is a steep decline from the first three. This highlights the strategies to avoid and lets us focus on farms that already have this kind of strategy in place. This could lead to investigations into what is happening, why the farmer is setting it up that way, or whether the farmer is even aware that the settings are set the way they are.

7. Personal Reflection

Looking back at my whole internship period, from start to finish, I think it has been a very pleasant and worthwhile experience. In terms of personal growth, I also improved technically in using my skills.

During this internship, I was exposed to SQL a lot, which I had not had much experience with aside from one other project. With this project being so SQL-heavy, it really helped me improve my SQL skills by a wide margin compared to before. Things like my proficiency in data manipulation and in working with/manipulating large, real-life, structured datasets really improved. I also improved my analytical skills; my ability to identify patterns in complex datasets improved, and, all in all, this internship gave me a better understanding of how data reflects real-world processes and how to translate real-world problems into data questions.

For my personal development, I also feel I have made some strides. Having finally experienced firsthand how the professional and corporate worlds work was very insightful and useful to see and live. Since it was my first time in a real professional environment, I had a lot to learn, and for now, this is only the start. However, I would say that, from this, I have gotten a pretty good start and understanding of this new era that many of my classmates and I will now enter in our careers.

Being able to work autonomously while also collaborating with teams of experts from different departments was useful for my development. Working cross-functionally with co-workers from other teams and specialties, translating my findings for nontechnical people,

and having meaningful conversations that led to useful feedback and advice I could take back for my analyses was an extremely useful experience I regularly practiced with them. But also having total freedom over my project and what I do is a point I cannot forget to mention, as it gave weight to my decisions and made me solely responsible for my work. Working under that kind of pretense was another useful experience to take back with me. And the combination of having autonomy over my work and having other colleagues to confide in, exchange questions and solutions with, is truly one of the more important skills I learned here.

7.1 Challenges and Limitations

The main challenge I faced during this internship was the Domain Knowledge hump. There was a lot of very technical, niche farming terminology and general knowledge that I had never been exposed to before coming to this company. This meant that a good portion of the start of this internship was spent trying to understand the kind of data I was working with and exposing myself to some general exploration of the data, so I could obtain the much-needed context to use this kind of data.

This was important because a good understanding of the data that one works with is crucial so that anything that was worked on, any results that I would receive, any methods I would use, could continue having validity and above all else, it made sure I stayed in the loop of what I was working with and made sure I maintained a good grip on the concepts of my work.

A lot of this process of understanding the domain-specific knowledge was about bridging my skill set and my world with this company's world, which included specific farming knowledge that, in most cases, no normal developer is going to be privy to. And this mattered because using my skills properly in a way that made sense and making sure I was not just wasting my time were good points to keep in mind. Several of my coworkers stated that it takes years to understand this kind of data, which startled me, but in the context of this project alone it was not so dire, and it did not require much deep knowledge.

This was addressed by consistently exposing myself to this kind of data every day and gradually learning my way through it. I also used various resources provided by colleagues about farming, dairy cattle, and other topics tied to my specific project context. And every time I was doubtful about the results or anything in general, I always validated them with the rest of the team to gain better insight into what to expect and a stronger understanding of the overall analytical meaning behind the numbers I was seeing.

7.2 Closing statement

Overall, the whole experience left me with a good impression of the future. The co-workers I had were all very nice and welcoming. The team was very diverse in every sense, and there is good support from the company, among other things.

For future development goals, I would like to explore and experience a different field of expertise within my domain. What exactly that is, I do not know yet, but I am always open to

exploring different kinds of offers that might be available to me. But in the same way, having the chance to improve upon the things I have learned during this period would also be important to me. Even if I had a long exposure now to the types of things I am doing and the tools that I am using, there is always something else for me to learn, and that openness to learn new things, new skills, new methods, new tools and new domains is what makes me grow above my potential and is all I could ask for as a person who is looking to grow each year.

References

1. European Commission. (2020, July 17). *Assessment list for trustworthy artificial intelligence (ALTAI) for self-assessment*. Shaping Europe's Digital Future. <https://digital-strategy.ec.europa.eu/en/library/assessment-list-trustworthy-artificial-intelligence-altai-self-assessment>
2. Ozella, L., Brotto Rebuli, K., Forte, C., & Giacobini, M. (2023). A literature review of modeling approaches applied to data collected in automatic milking systems. *Animals*, 13(12), 1916. <https://doi.org/10.3390/ani13121916>
3. Ji, B., Banhazi, T., Phillips, C.J.C., Wang, C., & Li, B. (2022). A machine learning framework to predict the next month's daily milk yield, milk composition and milking frequency for cows in a robotic dairy farm. *Biosystems Engineering*, 216, 186–197. <https://doi.org/10.1016/j.biosystemseng.2022.02.013>
4. Žgajnar, J. (2024). Strategic categorization of dairy cow farms in Croatia using cluster analysis. *Business Systems Research*, 15. <https://doi.org/10.2478/bsrj-2024-0021>
5. Atıl, H., & Akıllı, A. (2016). Comparison of artificial neural network and K-means for clustering dairy cattle. *International Journal of Sustainable Agricultural Management and Informatics*, 2(1), 40–52. <https://doi.org/10.1504/IJSAMI.2016.077266>
6. Brotzman, R. L., Cook, N. B., Nordlund, K., Bennett, T. B., Gomez Rivas, A., & Döpfer, D. (2015). Cluster analysis of Dairy Herd Improvement data to discover trends in performance characteristics in large Upper Midwest dairy herds. *Journal of Dairy Science*, 98(5), 3059–3070. <https://doi.org/10.3168/jds.2014-8949>
7. Ji, H., Hwang, I., Kim, J., Lee, S., & Lee, W. (2024). Leveraging feature extraction and risk-based clustering for advanced fault diagnosis in equipment. *PLOS ONE*, 19(12), e0314931. <https://doi.org/10.1371/journal.pone.0314931>
8. Lee, H., Choi, Y., Son, B., Lim, J., Lee, S., Kang, J. W., Kim, K. H., Kim, E. J., Yang, C., & Lee, J.-D. (2022). Deep autoencoder-powered pattern identification of sleep disturbance using multi-site cross-sectional survey data. *Frontiers in Medicine*, 9, 950327. <https://doi.org/10.3389/fmed.2022.950327>
9. Matzhöld, C., Schodl, K., Klimek, P., Steininger, F., & Egger-Danner, C. (2024). A key-feature-based clustering approach to assess the impact of technology integration on cow health in Austrian dairy farms. *Frontiers in Animal Science*, 5, 1421299. <https://doi.org/10.3389/fanim.2024.1421299>
10. Riaz, M. U., O'Grady, L., McAloon, C. G., Logan, F., & Gormley, I. C. (2025). Comparison of machine learning and validation methods for high-dimensional accelerometer data to detect foot lesions in dairy cattle. *PLOS ONE*, 20(6), e0325927. <https://doi.org/10.1371/journal.pone.0325927>
11. Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
12. Michelena, Á., Díaz-Longueira, A., Novais, P., Simić, D., Fontenla-Romero, Ó., & Calvo-Rolle, J. L. (2025). Comparative analysis of unsupervised anomaly detection techniques for heat detection in dairy cattle. *Neurocomputing*, 618, 129088. <https://doi.org/10.1016/j.neucom.2024.129088>

13. Trapanese, L., Bifulco, G., Santinello, M., Pasquino, N., Campanile, G., & Salzano, A. (2025). Comparison of K-means and hierarchical clustering methods for buffalo milk production data. *Animals*, 15(22), 3246.
<https://doi.org/10.3390/ani15223246>
14. Thorndike, R. L. (1953). Who belongs in the family? *Psychometrika*, 18(4), 267–276. <https://doi.org/10.1007/BF02289263>
15. Microsoft. (2026). *M365 Copilot (GPT-5 chat model)* [Large language model].
<https://copilot.microsoft.com>